

ONLINE RETAIL DATA ANALYSIS USING HADOOP



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Table of Contents:

Sr.No.	Topic	Page No.
1.	Problem Statement	
2.	Dataset Description	
3.	Summary of Data Analysis	
4.	Map Reduce Analysis	
5.	Pig Analysis	

Problem Statement

Analyse the UCI Online Retail transaction dataset using Hadoop MapReduce and Apache Pig, to determine which products, customers and countries generate the most revenue and how sales behave over time.

Dataset Description

Dataset: Online Retail, UCI Machine Learning Repository

Source: <https://archive.ics.uci.edu/dataset/352/online+retail>

The dataset records every transaction of a UK-based online gift retailer between 1 December 2010 and 9 December 2011, a total of 541,909 rows. The publisher distributes it as a single Excel workbook; because Hadoop reads plain text, the workbook was converted to CSV and partitioned by invoice month into 13 files, so that HDFS produces 13 input splits and the jobs run 13 map tasks in parallel.

InvoiceNo	invoice number; a leading C marks a cancelled order
StockCode	product identifier
Description	product name
Quantity	units sold; negative on returns
InvoiceDate	date and time of the transaction
UnitPrice	price per unit in pounds sterling
CustomerID	customer identifier; empty on 24.9% of rows
Country	country of the customer

Preparation: commas inside Description were replaced with spaces (4,796 rows) and quote characters removed (1,239 rows), so that every field splits cleanly on a comma. Country contains no commas. After conversion all 541,909 rows split into exactly eight fields.

Summary of Data Analysis

The following MapReduce design patterns are implemented:

1. Filtering
2. Summarization
3. Data Organization
4. Top-N selection

The following MapReduce analyses are performed:

1. Total sales revenue by product
2. Average invoice value, with per-invoice detail
3. Top 10 customers by total spending
4. Binning of transactions by country

The following Pig analyses are performed:

1. Top 5 countries by revenue
2. Top 10 products by revenue
3. Top 10 products in each country
4. Top 10 customers by spending
5. Monthly sales trend

Map Reduce Analysis

Total Sales by Product

Returns the total revenue generated by each product, calculated as Quantity multiplied by UnitPrice and summed per StockCode. Cancelled invoices are excluded.

1. Driver Class : sets up configuration for the Mapper, Combiner and Reducer
2. Mapper Class : emits StockCode of type Text and line revenue of type DoubleWritable
3. Reducer Class : emits StockCode as key and its total revenue of type DoubleWritable

Driver Class

```
1 package totalsalesbyproduct;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.conf.Configuration;
6 import org.apache.hadoop.fs.Path;
7 import org.apache.hadoop.io.DoubleWritable;
8 import org.apache.hadoop.io.Text;
9 import org.apache.hadoop.mapreduce.Job;
10 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
11 import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
12 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
13 import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
14
15 /**
16  * Total Sales by Product : returns the total revenue generated by every product,
17  * ignoring cancelled invoices. Sets up configuration for the Mapper, Combiner
18  * and Reducer classes.
19  */
20 public class DriverClass {
21
22     public static void main(String[] args)
23         throws IOException, InterruptedException, ClassNotFoundException {
24
```

```
25     Configuration conf = new Configuration();
26     Job job = Job.getInstance(conf, "Total Sales By Product");
27     job.setJarByClass(DriverClass.class);
28
29     job.setMapperClass(MapperClass.class);
30     job.setCombinerClass(ReducerClass.class);
31     job.setReducerClass(ReducerClass.class);
32
33     job.setMapOutputKeyClass(Text.class);
34     job.setMapOutputValueClass(DoubleWritable.class);
35     job.setOutputKeyClass(Text.class);
36     job.setOutputValueClass(DoubleWritable.class);
37
38     job.setInputFormatClass(TextInputFormat.class);
39     job.setOutputFormatClass(TextOutputFormat.class);
40
41     FileInputFormat.addInputPath(job, new Path(args[0]));
42     FileOutputFormat.setOutputPath(job, new Path(args[1]));
43
44     System.exit(job.waitForCompletion(true) ? 0 : 1);
45 }
46 }
```

Mapper Class

```
1 package totalsalesbyproduct;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.DoubleWritable;
6 import org.apache.hadoop.io.LongWritable;
7 import org.apache.hadoop.io.Text;
8 import org.apache.hadoop.mapreduce.Mapper;
9
10 /**
11  * Emits StockCode of type Text and the line revenue (Quantity * UnitPrice) of
12  * type DoubleWritable.
13  *
14  * Dataset columns: 0 InvoiceNo, 1 StockCode, 2 Description, 3 Quantity,
15  * 4 InvoiceDate, 5 UnitPrice, 6 CustomerID, 7 Country.
16  */
17 public class MapperClass extends Mapper<LongWritable, Text, Text, DoubleWritable> {
18
19     private Text stockCode = new Text();
20     private DoubleWritable revenue = new DoubleWritable();
21
22     @Override
23     public void map(LongWritable key, Text value, Context context)
24         throws IOException, InterruptedException {
```

```
25
26     String[] col = value.toString().split(",");
27     if (col.length < 8) {
28         return;
29     }
30
31     // An InvoiceNo beginning with C marks a cancelled order; those rows carry
32     // negative quantities and would subtract from genuine sales.
33     if (col[0].startsWith("C") || col[0].startsWith("c")) {
34         context.getCounter("Retail", "CancelledRowsSkipped").increment(1);
35         return;
36     }
37
38     try {
39         double qty = Double.parseDouble(col[3]);
40         double price = Double.parseDouble(col[5]);
41         if (qty <= 0 || price <= 0) {
42             context.getCounter("Retail", "NonPositiveRowsSkipped").increment(1);
43             return;
44         }
45         stockCode.set(col[1]);
46         revenue.set(qty * price);
47         context.write(stockCode, revenue);
48     } catch (NumberFormatException e) {
49         context.getCounter("Retail", "UnparseableRowsSkipped").increment(1);
50     }
51 }
52 }
```

Reducer Class

```
1 package totalsalesbyproduct;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.DoubleWritable;
6 import org.apache.hadoop.io.Text;
7 import org.apache.hadoop.mapreduce.Reducer;
8
9 /**
10  * Emits StockCode as key of type Text and its total revenue of type
11  * DoubleWritable. Also usable as the Combiner, since summing is associative.
12  */
13 public class ReducerClass extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
14
15     private DoubleWritable total = new DoubleWritable();
16
17     @Override
18     public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
19         throws IOException, InterruptedException {
20
21         double sum = 0;
22         for (DoubleWritable value : values) {
23             sum += value.get();
24         }
25         total.set(Math.round(sum * 100.0) / 100.0);
26         context.write(key, total);
27     }
28 }
```

Execution

```
YTRUN-02_total_sales_by_product

D:\Desktop\bda1_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib\* -d D:\Desktop\bda1_project\TotalSalesByProduct\target\classes D:\Desktop\bda1_project\TotalSalesByProduct\src\main\java\totalsalesbyproduct\DriverClass.java D:\Desktop\bda1_project\TotalSalesByProduct\src\main\java\totalsalesbyproduct\MapperClass.java D:\Desktop\bda1_project\TotalSalesByProduct\src\main\java\totalsalesbyproduct\ReducerClass.java

D:\Desktop\bda1_project>jar -cvf D:\Desktop\bda1_project\TotalSalesByProduct\target\TotalSalesByProduct.jar -C D:\Desktop\bda1_project\TotalSalesByProduct\target\classes .
added manifest
adding: totalsalesbyproduct/(in = 0) (out= 0)(stored 0%)
adding: totalsalesbyproduct/DriverClass.class(in = 1843) (out= 918)(deflated 50%)
adding: totalsalesbyproduct/MapperClass.class(in = 2471) (out= 1081)(deflated 56%)
adding: totalsalesbyproduct/ReducerClass.class(in = 1798) (out= 776)(deflated 56%)

D:\Desktop\bda1_project>hadoop jar D:\Desktop\bda1_project\TotalSalesByProduct\target\TotalSalesByProduct.jar totalsalesbyproduct.DriverClass /Bda1Project/data /Bda1Project/output/totalsales
2026-09-08 17:42:33,567 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:42:34,722 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:42:34,738 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/HP/.staging/job_1788866353307_0007
2026-09-08 17:42:35,027 INFO input.FileInputFormat: Total input files to process : 13
2026-09-08 17:42:35,125 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:42:35,287 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0007
2026-09-08 17:42:35,291 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:42:35,509 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:42:35,510 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:42:35,603 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0007
2026-09-08 17:42:35,667 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/application_1788866353307_0007/
2026-09-08 17:42:35,669 INFO mapreduce.Job: Running job: job_1788866353307_0007
2026-09-08 17:42:53,028 INFO mapreduce.Job: Job job_1788866353307_0007 running in uber mode : false
2026-09-08 17:42:53,029 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:43:15,783 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:43:18,993 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:43:21,117 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:43:26,417 INFO mapreduce.Job: map 38% reduce 0%
2026-09-08 17:43:27,466 INFO mapreduce.Job: map 46% reduce 0%
2026-09-08 17:43:30,522 INFO mapreduce.Job: map 54% reduce 0%
2026-09-08 17:43:47,009 INFO mapreduce.Job: map 62% reduce 0%
2026-09-08 17:43:48,035 INFO mapreduce.Job: map 69% reduce 0%
2026-09-08 17:43:50,107 INFO mapreduce.Job: map 77% reduce 0%
```

```
YTRUN-02_total_sales_by_product

Reduce input records=34069
Reduce output records=3922
Spilled Records=68138
Shuffled Maps =13
Failed Shuffles=0
Merged Map outputs=13
GC time elapsed (ms)=2217
CPU time spent (ms)=35101
Physical memory (bytes) snapshot=4473888768
Virtual memory (bytes) snapshot=6052175872
Total committed heap usage (bytes)=3779592192
Peak Map Physical memory (bytes)=343986176
Peak Map Virtual memory (bytes)=450805760
Peak Reduce Physical memory (bytes)=219717632
Peak Reduce Virtual memory (bytes)=322060288

Retail
CancelledRowsSkipped=9288
NonPositiveRowsSkipped=2517
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=45458480
File Output Format Counters
Bytes Written=52300

D:\Desktop\bda1_project>hdfs dfs -ls /Bda1Project/output/totalsales
Found 2 items
-rw-r--r-- 1 HP supergroup 0 2026-09-08 17:43 /Bda1Project/output/totalsales/_SUCCESS
-rw-r--r-- 1 HP supergroup 52300 2026-09-08 17:43 /Bda1Project/output/totalsales/part-r-00000

D:\Desktop\bda1_project>hdfs dfs -head /Bda1Project/output/totalsales/part-r-00000
10002 759.89
10080 119.09
10120 40.53
10123C 3.25
10124A 6.72
10124G 7.14
10125 994.84
```

YTRUN-02_total_sales_by_product_out

D:\Desktop\bdal_project>hdfs dfs -head /BdalProject/output/totalsales/part-r-00000

10002	759.89
10080	119.09
10120	40.53
10123C	3.25
10124A	6.72
10124G	7.14
10125	994.84
10133	1544.22
10135	2207.39
11001	2389.44
15030	41.47
15034	840.02
15036	18569.84
15039	1979.49
15044A	1456.72
15044B	946.83
15044C	1025.94
15044D	1822.71
15056BL	15828.84
15056N	22176.49
15056P	4366.82
15056bl	1086.88
15056n	1585.28
15056p	412.48
15058A	1668.49
15058B	1885.27
15058C	1429.82
15060B	1807.55
15060b	24.87
16008	360.71
16010	3.6
16011	372.33
16012	261.24
16014	4335.76
16015	289.98
16016	1053.45
16020C	69.9
16033	100.8
16043	11.76
16045	380.0
16046	57.8

YTRUN-02_total_sales_by_product_out	
16046	57.8
16048	205.42
16049	82.08
16052	115.3
16054	140.52
16151A	61.25
16156L	199.5
16156S	2331.0
16161G	70.0
16161M	10.5
16161P	2990.92
16161U	1687.5
16162L	13.0
16162M	11.76
16168M	173.12
16169E	1892.46
16169K	84.0
16169M	277.2
16169N	10.5
16169P	21.0
16202A	3.5
16202B	14.5
16202E	7.5
16206B	8.75
16207A	163.1
16207B	76.14
16216	234.0
16218	283.31
16219	407.72
16225	691.85
16235	702.96
16236	320.09
16237	1048.56
16238	322.09
16244B	7.85
16248B	16.25
16254	42.58
16258A	1

Average Invoice Value

Returns the number of order lines, the total value and the mean line value of every invoice, together with the average invoice value across the whole dataset.

1. Driver Class : sets up configuration for the Mapper, Combiner, Reducer and Tuple class
2. Mapper Class : emits InvoiceNo of type Text and a Tuple of custom type InvoiceValueTuple
3. Combiner Class : merges the order lines of one invoice on the map side
4. Reducer Class : emits InvoiceNo as key and the summarised Tuple, and writes the overall average from cleanup()
5. Tuple Class : POJO carrying the line count and their summed value

Driver Class

```
1 package averageinvoicevalue;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.conf.Configuration;
6 import org.apache.hadoop.fs.Path;
7 import org.apache.hadoop.io.Text;
8 import org.apache.hadoop.mapreduce.Job;
9 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
10 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
11
12 /**
13  * Average Invoice Value : returns the line count, total value and mean line
14  * value of every invoice, and the average invoice value across the whole
15  * dataset. Sets up configuration for the Mapper, Combiner, Reducer and Tuple
16  * class.
17  */
18 public class DriverClass {
19
20     public static void main(String[] args)
21         throws IOException, InterruptedException, ClassNotFoundException {
22
23         Configuration conf = new Configuration();
24         Job job = Job.getInstance(conf, "Average Invoice Value");
```

```

25     job.setJarByClass(DriverClass.class);
26
27     job.setMapperClass(MapperClass.class);
28     job.setCombinerClass(CombinerClass.class);
29     job.setReducerClass(ReducerClass.class);
30
31     job.setMapOutputKeyClass(Text.class);
32     job.setMapOutputValueClass(InvoiceValueTuple.class);
33     job.setOutputKeyClass(Text.class);
34     job.setOutputValueClass(InvoiceValueTuple.class);
35
36     // A single reducer, so the dataset-wide average written from cleanup()
37     // is computed once over every invoice rather than once per partition.
38     job.setNumReduceTasks(1);
39
40     FileInputFormat.addInputPath(job, new Path(args[0]));
41     FileOutputFormat.setOutputPath(job, new Path(args[1]));
42
43     System.exit(job.waitForCompletion(true) ? 0 : 1);
44 }
45 }

```

Mapper Class

```

1  package averageinvoicevalue;
2
3  import java.io.IOException;
4
5  import org.apache.hadoop.io.LongWritable;
6  import org.apache.hadoop.io.Text;
7  import org.apache.hadoop.mapreduce.Mapper;
8
9  /**
10   * Emits InvoiceNo of type Text and a Tuple of custom type InvoiceValueTuple
11   * holding one order line and its value.
12   */
13  public class MapperClass extends Mapper<LongWritable, Text, Text, InvoiceValueTuple> {
14
15     private Text invoiceNo = new Text();
16     private InvoiceValueTuple outTuple = new InvoiceValueTuple();
17
18     @Override
19     public void map(LongWritable key, Text value, Context context)
20         throws IOException, InterruptedException {
21
22         String[] col = value.toString().split(",");
23         if (col.length < 8) {
24             return;

```

```
25     }
26     if (col[0].startsWith("C") || col[0].startsWith("c")) {
27         context.getCounter("Retail", "CancelledRowsSkipped").increment(1);
28         return;
29     }
30
31     try {
32         double qty = Double.parseDouble(col[3]);
33         double price = Double.parseDouble(col[5]);
34         if (qty <= 0 || price <= 0) {
35             return;
36         }
37         invoiceNo.set(col[0]);
38         outTuple.setLineCount(1);
39         outTuple.setTotalAmount(qty * price);
40         context.write(invoiceNo, outTuple);
41     } catch (NumberFormatException e) {
42         context.getCounter("Retail", "UnparseableRowsSkipped").increment(1);
43     }
44 }
45 }
```

Combiner Class

```
1 package averageinvoicevalue;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7
8 /**
9  * Combiner Class : merges the order lines of one invoice on the map side.
10  * Line count and amount are both carried forward so the reducer still sees a
11  * faithful partial sum.
12  */
13 public class CombinerClass extends Reducer<Text, InvoiceValueTuple, Text, InvoiceValueTuple> {
14
15     private InvoiceValueTuple result = new InvoiceValueTuple();
16
17     @Override
18     protected void reduce(Text key, Iterable<InvoiceValueTuple> values, Context context)
19         throws IOException, InterruptedException {
20
21         int lines = 0;
22         double amount = 0;
23         for (InvoiceValueTuple val : values) {
24             lines += val.getLineCount();
25             amount += val.getTotalAmount();
26         }
27         result.setLineCount(lines);
28         result.setTotalAmount(amount);
29         context.write(key, result);
30     }
31 }
```

Reducer Class

```
1 package averageinvoicevalue;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7
8 /**
9  * Emits InvoiceNo as key of type Text and a Tuple of custom type
10  * InvoiceValueTuple giving the line count, invoice total and mean line value.
11  *
12  * The overall average invoice value cannot be produced per key, so the running
13  * totals are held across reduce() calls and written once from cleanup().
14  */
15 public class ReducerClass extends Reducer<Text, InvoiceValueTuple, Text, InvoiceValueTuple> {
16
17     private InvoiceValueTuple result = new InvoiceValueTuple();
18     private int invoiceCount = 0;
19     private double grandTotal = 0;
20
21     @Override
22     protected void reduce(Text key, Iterable<InvoiceValueTuple> values, Context context)
23         throws IOException, InterruptedException {
24
```

```
25     int lines = 0;
26     double amount = 0;
27     for (InvoiceValueTuple val : values) {
28         lines += val.getLineCount();
29         amount += val.getTotalAmount();
30     }
31     result.setLineCount(lines);
32     result.setTotalAmount(amount);
33     context.write(key, result);
34
35     invoiceCount++;
36     grandTotal += amount;
37 }
38
39 @Override
40 protected void cleanup(Context context) throws IOException, InterruptedException {
41     if (invoiceCount == 0) {
42         return;
43     }
44     InvoiceValueTuple overall = new InvoiceValueTuple();
45     overall.setLineCount(invoiceCount);
46     overall.setTotalAmount(grandTotal);
47     context.write(new Text("ZZZ_AVERAGE_INVOICE_VALUE"), overall);
48 }
49 }
```

Tuple Class

```
1 package averageinvoicevalue;
2
3 import java.io.DataInput;
4 import java.io.DataOutput;
5 import java.io.IOException;
6
7 import org.apache.hadoop.io.Writable;
8
9 /**
10  * Tuple Class : POJO carrying the number of order lines and their summed value.
11  * Both parts must travel together, otherwise the Combiner cannot merge partial
12  * results without distorting the average.
13  */
14 public class InvoiceValueTuple implements Writable {
15
16     private int lineCount = 0;
17     private double totalAmount = 0;
18
19     public int getLineCount() {
20         return lineCount;
21     }
22
23     public void setLineCount(int lineCount) {
24         this.lineCount = lineCount;
```

```
25     }
26
27     public double getTotalAmount() {
28         return totalAmount;
29     }
30
31     public void setTotalAmount(double totalAmount) {
32         this.totalAmount = totalAmount;
33     }
34
35     public void write(DataOutput d) throws IOException {
36         d.writeInt(lineCount);
37         d.writeDouble(totalAmount);
38     }
39
40     public void readFields(DataInput di) throws IOException {
41         lineCount = di.readInt();
42         totalAmount = di.readDouble();
43     }
44
45     @Override
46     public String toString() {
47         double avg = (lineCount == 0) ? 0 : totalAmount / lineCount;
48         return lineCount + "\t"
49             + (Math.round(totalAmount * 100.0) / 100.0) + "\t"
50             + (Math.round(avg * 100.0) / 100.0);
51     }
52 }
```

Execution

```
YTRUN-03_average_invoice_value

D:\Desktop\bda1_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib\* -d D:\Desktop\bda1_project\AverageInvoiceValue\target\classes D:\Desktop\bda1_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\DriverClass.java D:\Desktop\bda1_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\MapperClass.java D:\Desktop\bda1_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\CombinerClass.java D:\Desktop\bda1_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\ReducerClass.java D:\Desktop\bda1_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\InvoiceValueTuple.java

D:\Desktop\bda1_project>jar -cvf D:\Desktop\bda1_project\AverageInvoiceValue\target\AverageInvoiceValue.jar -C D:\Desktop\bda1_project\AverageInvoiceValue\target\classes .
added manifest
adding: averageinvoicevalue/(in = 0) (out= 0)(stored 0%)
adding: averageinvoicevalue/CombinerClass.class(in = 1869) (out= 801)(deflated 57%)
adding: averageinvoicevalue/DriverClass.class(in = 1717) (out= 892)(deflated 48%)
adding: averageinvoicevalue/InvoiceValueTuple.class(in = 1478) (out= 804)(deflated 45%)
adding: averageinvoicevalue/MapperClass.class(in = 2495) (out= 1100)(deflated 55%)
adding: averageinvoicevalue/ReducerClass.class(in = 2435) (out= 1018)(deflated 58%)

D:\Desktop\bda1_project>hadoop jar D:\Desktop\bda1_project\AverageInvoiceValue\target\AverageInvoiceValue.jar averageinvoicevalue.DriverClass /Bda1Project/data /Bda1Project/output/avginvoice
2026-09-08 17:46:09,364 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:46:10,859 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:46:10,875 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/HP/.staging/job_1788866353307_0008
2026-09-08 17:46:11,412 INFO input.FileInputFormat: Total input files to process : 13
2026-09-08 17:46:11,715 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:46:12,350 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0008
2026-09-08 17:46:12,373 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:46:13,163 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:46:13,165 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:46:13,276 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0008
2026-09-08 17:46:13,346 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/application_1788866353307_0008/
2026-09-08 17:46:13,347 INFO mapreduce.Job: Running job: job_1788866353307_0008
2026-09-08 17:46:34,824 INFO mapreduce.Job: Job job_1788866353307_0008 running in uber mode : false
2026-09-08 17:46:34,826 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:46:57,468 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:47:01,601 INFO mapreduce.Job: map 15% reduce 0%
2026-09-08 17:47:04,747 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:47:05,948 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:47:07,051 INFO mapreduce.Job: map 38% reduce 0%
```

YTRUN-03_average_invoice_value			
536367	12	278.73	23.23
536368	4	70.05	17.51
536369	1	17.85	17.85
536370	20	855.86	42.79
536371	1	204.0	204.0
536372	2	22.2	11.1
536373	16	259.86	16.24
536374	1	350.4	350.4
536375	16	259.86	16.24
536376	2	328.8	164.4
536377	2	22.2	11.1
536378	19	444.98	23.42
536380	1	34.8	34.8
536381	35	449.98	12.86
536382	12	430.6	35.88
536384	13	489.6	37.66
536385	7	130.85	18.69
536386	3	508.2	169.4
536387	5	3193.92	638.78
536388	14	226.14	16.15
536389	14	358.25	25.59
536390	24	1825.74	76.07
536392	10	318.14	31.81
536393	1	79.6	79.6
536394	11	1024.68	93.15
536395	14	507.88	36.28
536396	18	376.36	20.91
536397	2	279.0	139.5
536398	17	426.56	25.09
536399	2	22.2	11.1
536400	1	17.4	17.4
536401	64	354.23	5.53
536402	3	357.0	119.0
536403	2	192.6	96.3
536404	28	471.3	16.83
536405	1	326.4	326.4
536406	17	353.14	20.77
536407	2	22.2	11.1
536408	48	783.11	16.31
536409	58	243.28	4.19
536412	81	514.41	6.35
536415	59	390.79	6.62
536416	6	225.7	37.62

```
YTRUN-03_average_invoice_value_out
D:\Desktop\bda1_project>hdfs dfs -tail /Bda1Project/output/avginvoice/part-r-00000
2618.89 40.29
581477 10      291.78  29.18
581478 18      391.02  21.72
581479 1       29.5    29.5
581480 9        305.7   33.97
581481 16       242.3   15.14
581482 6        488.4   81.4
581483 1       168469.6 168469.6
581485 3        741.9   247.3
581486 16       407.68  25.48
581487 1       406.8   406.8
581488 3        415.6   138.53
581489 2       171.6   85.8
581491 1       19.8    19.8
581492 731     7689.23 10.52
581493 17      208.64  12.27
581494 18      339.14  18.84
581495 15     1203.9  80.26
581496 32      598.97  18.72
581497 59     3217.2  54.53
581498 218    5664.89 25.99
581500 3       103.3   34.43
581501 24      485.0   20.21
581502 20     133.35  6.67
581514 24     200.99  8.37
581516 15     382.01  25.47
581538 74     197.09  2.66
581566 1       469.44  469.44
581567 12     389.21  32.43
581570 13     224.95  17.3
581571 20     494.6   24.73
581572 4       250.52  62.63
581574 13     277.08  21.31
581578 38     848.55  22.33
581579 30     903.62  30.12
581580 24     94.11   3.92
581581 3       51.18   17.06
581582 2       29.88   14.94
581583 2       124.6   62.3
581584 2      140.64  70.32
581585 21     329.05  15.67
```

```
YTRUN-03_average_invoice_value_out
581585 21     329.05  15.67
581586 4       339.2   84.8
581587 15     249.45  16.63
A563185 1     11062.06 11062.06
ZZZ_AVERAGE_INVOICE_VALUE 19960 1.066668454E7 534.4
```

Top 10 Customers by Spending

Returns the ten customers with the highest total spend, in descending order. Rows carrying no CustomerID are anonymous till transactions and are excluded.

1. Mapper Class : emits CustomerID of type Text and line revenue of type DoubleWritable
2. Reducer Class : totals each customer and retains only the ten highest in a TreeMap, written from cleanup()
3. Driver Class : sets a single reducer so one ranking is produced

Code

```
1 package topcustomersbyspending;
2
3 import java.io.IOException;
4 import java.util.Map;
5 import java.util.TreeMap;
6
7 import org.apache.hadoop.conf.Configuration;
8 import org.apache.hadoop.fs.Path;
9 import org.apache.hadoop.io.DoubleWritable;
10 import org.apache.hadoop.io.LongWritable;
11 import org.apache.hadoop.io.Text;
12 import org.apache.hadoop.mapreduce.Job;
13 import org.apache.hadoop.mapreduce.Mapper;
14 import org.apache.hadoop.mapreduce.Reducer;
15 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
16 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
17
18 /**
19  * Top 10 Customers by Spending : returns the ten customers with the highest
20  * total spend, in descending order.
21  *
22  * 135,080 of the 541,909 rows carry no CustomerID; those are anonymous
23  * till transactions and are excluded, since they cannot be attributed.
24  */
```

```

25 public class TopCustomersBySpending {
26
27     public static final int TOP_N = 10;
28
29     public static class Map1 extends Mapper<LongWritable, Text, Text, DoubleWritable> {
30
31         private Text customer = new Text();
32         private DoubleWritable spend = new DoubleWritable();
33
34         @Override
35         public void map(LongWritable key, Text value, Context context)
36             throws IOException, InterruptedException {
37
38             String[] col = value.toString().split(",");
39             if (col.length < 8) {
40                 return;
41             }
42             if (col[0].startsWith("C") || col[0].startsWith("c")) {
43                 context.getCounter("Retail", "CancelledRowsSkipped").increment(1);
44                 return;
45             }
46             if (col[6] == null || col[6].trim().isEmpty()) {
47                 context.getCounter("Retail", "NoCustomerIdSkipped").increment(1);
48                 return;

```

```

49         }
50
51         try {
52             double qty = Double.parseDouble(col[3]);
53             double price = Double.parseDouble(col[5]);
54             if (qty <= 0 || price <= 0) {
55                 return;
56             }
57             customer.set(col[6].trim());
58             spend.set(qty * price);
59             context.write(customer, spend);
60         } catch (NumberFormatException e) {
61             context.getCounter("Retail", "UnparseableRowsSkipped").increment(1);
62         }
63     }
64 }
65
66 public static class Reduce1 extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
67
68     // Ordered by spend; only the top N are ever retained, so memory stays bounded
69     // no matter how many customers the dataset holds.
70     private TreeMap<Double, String> ranked = new TreeMap<Double, String>();
71
72     @Override

```

```

73     public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
74         throws IOException, InterruptedException {
75
76         double sum = 0;
77         for (DoubleWritable value : values) {
78             sum += value.get();
79         }
80         ranked.put(sum, key.toString());
81         if (ranked.size() > TOP_N) {
82             ranked.remove(ranked.firstKey());
83         }
84     }
85
86     @Override
87     protected void cleanup(Context context) throws IOException, InterruptedException {
88         for (Map.Entry<Double, String> e : ranked.descendingMap().entrySet()) {
89             context.write(new Text(e.getValue()),
90                 new DoubleWritable(Math.round(e.getKey() * 100.0) / 100.0));
91         }
92     }
93 }
94
95 public static void main(String[] args)
96     throws IOException, InterruptedException, ClassNotFoundException {

```

```

97
98     Configuration conf = new Configuration();
99     Job job = Job.getInstance(conf, "Top 10 Customers By Spending");
100     job.setJarByClass(TopCustomersBySpending.class);
101
102     job.setMapperClass(Map1.class);
103     job.setReducerClass(Reduce1.class);
104
105     job.setMapOutputKeyClass(Text.class);
106     job.setMapOutputValueClass(DoubleWritable.class);
107     job.setOutputKeyClass(Text.class);
108     job.setOutputValueClass(DoubleWritable.class);
109
110     // One reducer, so a single ranking is produced rather than one per partition.
111     job.setNumReduceTasks(1);
112
113     FileInputFormat.addInputPath(job, new Path(args[0]));
114     FileOutputFormat.setOutputPath(job, new Path(args[1]));
115
116     System.exit(job.waitForCompletion(true) ? 0 : 1);
117 }
118 }

```

Execution

```
YTRUN-04_top_customers_by_spending

D:\Desktop\bdal_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\
*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib\* -d D:\Desktop\bdal_project\TopCustomer
sBySpending\target\classes D:\Desktop\bdal_project\TopCustomersBySpending\src\main\java\topcustomersbyspen
ding\TopCustomersBySpending.java

D:\Desktop\bdal_project>jar -cvf D:\Desktop\bdal_project\TopCustomersBySpending\target\TopCustomersBySpending.jar -C D:\Desktop\bdal_project\TopCustomersBySpending\target\classes .
added manifest
adding: topcustomersbyspending/(in = 0) (out= 0)(stored 0%)
adding: topcustomersbyspending/TopCustomersBySpending$Map1.class(in = 2641) (out= 1151)(deflated 56%)
adding: topcustomersbyspending/TopCustomersBySpending$Reduce1.class(in = 3031) (out= 1248)(deflated 58%)
adding: topcustomersbyspending/TopCustomersBySpending.class(in = 1814) (out= 943)(deflated 48%)

D:\Desktop\bdal_project>hadoop jar D:\Desktop\bdal_project\TopCustomersBySpending\target\TopCustomersBySpending.jar topcustomersbyspending.TopCustomersBySpending /BdalProject/data /BdalProject/output/topcustomers
2026-09-08 17:48:43,495 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:48:44,961 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:48:44,982 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
-yarn/staging/HP/.staging/job_1788866353307_0009
2026-09-08 17:48:45,347 INFO input.FileInputFormat: Total input files to process : 13
2026-09-08 17:48:45,473 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:48:45,655 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0009
2026-09-08 17:48:45,658 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:48:45,950 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:48:45,951 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:48:46,045 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0009
2026-09-08 17:48:46,117 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/ap
plication_1788866353307_0009/
2026-09-08 17:48:46,118 INFO mapreduce.Job: Running job: job_1788866353307_0009
2026-09-08 17:49:03,426 INFO mapreduce.Job: Job job_1788866353307_0009 running in uber mode : false
2026-09-08 17:49:03,430 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:49:22,981 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:49:27,169 INFO mapreduce.Job: map 15% reduce 0%
2026-09-08 17:49:32,384 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:49:33,428 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:49:35,485 INFO mapreduce.Job: map 38% reduce 0%
2026-09-08 17:49:38,572 INFO mapreduce.Job: map 46% reduce 0%
2026-09-08 17:49:41,626 INFO mapreduce.Job: map 54% reduce 0%
2026-09-08 17:49:53,876 INFO mapreduce.Job: map 62% reduce 0%
2026-09-08 17:49:56,950 INFO mapreduce.Job: map 69% reduce 0%
2026-09-08 17:50:01,053 INFO mapreduce.Job: map 77% reduce 0%
```

```
YTRUN-04_top_customers_by_spending

Reduce input records=397884
Reduce output records=10
Spilled Records=795768
Shuffled Maps =13
Failed Shuffles=0
Merged Map outputs=13
GC time elapsed (ms)=2030
CPU time spent (ms)=33239
Physical memory (bytes) snapshot=4511465472
Virtual memory (bytes) snapshot=5971415040
Total committed heap usage (bytes)=3747086336
Peak Map Physical memory (bytes)=340942848
Peak Map Virtual memory (bytes)=450437120
Peak Reduce Physical memory (bytes)=222162944
Peak Reduce Virtual memory (bytes)=309403648

Retail
CancelledRowsSkipped=9288
NoCustomerIdSkipped=134697
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=45458480
File Output Format Counters
Bytes Written=154

D:\Desktop\bda1_project>hdfs dfs -ls /Bda1Project/output/topcustomers
Found 2 items
-rw-r--r-- 1 HP supergroup 0 2026-09-08 17:50 /Bda1Project/output/topcustomers/_SUCCESS
-rw-r--r-- 1 HP supergroup 154 2026-09-08 17:50 /Bda1Project/output/topcustomers/part-r-00000

D:\Desktop\bda1_project>hdfs dfs -cat /Bda1Project/output/topcustomers/part-r-00000
14646 280206.02
18102 259657.3
17450 194550.79
16446 168472.5
14911 143825.06
12415 124914.53
14156 117379.63
```

```
YTRUN-04_top_customers_by_spending_out

D:\Desktop\bda1_project>hdfs dfs -cat /Bda1Project/output/topcustomers/part-r-00000
14646 280206.02
18102 259657.3
17450 194550.79
16446 168472.5
14911 143825.06
12415 124914.53
14156 117379.63
17511 91062.38
16029 81024.84
12346 77183.6
-
```

Binning by Country

Splits the transactions into one output file per country using MultipleOutputs. No aggregation is required, only routing, so the reduce phase is switched off and the job is map-only.

1. Driver Class : sets up configuration for the Mapper and registers the named output

2. Mapper Class : uses MultipleOutputs to write each row to the file for its country

Code

```
1 package binningbycountry;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.conf.Configuration;
6 import org.apache.hadoop.fs.Path;
7 import org.apache.hadoop.io.LongWritable;
8 import org.apache.hadoop.io.NullWritable;
9 import org.apache.hadoop.io.Text;
10 import org.apache.hadoop.mapreduce.Job;
11 import org.apache.hadoop.mapreduce.Mapper;
12 import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
13 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
14 import org.apache.hadoop.mapreduce.lib.output.MultipleOutputs;
15 import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
16
17 /**
18  * Binning by Country : splits the transactions into one output file per
19  * country using MultipleOutputs. This is a map-only job - no aggregation is
20  * required, only routing - so the reduce phase is switched off entirely.
21  */
22 public class BinningByCountry {
23
24     public static class BinMapper extends Mapper<LongWritable, Text, Text, NullWritable> {
```

```

25
26     private MultipleOutputs<Text, NullWritable> mos = null;
27
28     @Override
29     protected void setup(Context context) throws IOException, InterruptedException {
30         mos = new MultipleOutputs<Text, NullWritable>(context);
31     }
32
33     @Override
34     protected void map(LongWritable key, Text value, Context context)
35         throws IOException, InterruptedException {
36
37         String[] col = value.toString().split(",");
38         if (col.length < 8) {
39             return;
40         }
41         String country = col[7].trim();
42         if (country.isEmpty()) {
43             country = "Unspecified";
44         }
45         // The bin name becomes part of a file name, so anything that is not a
46         // letter or digit is folded to an underscore.
47         String bin = country.replaceAll("[^A-Za-z0-9]", "_");
48         mos.write("bins", value, NullWritable.get(), bin);

```

```

49         context.getCounter("Country", country).increment(1);
50     }
51
52     @Override
53     protected void cleanup(Context context) throws IOException, InterruptedException {
54         mos.close();
55     }
56 }
57
58 public static void main(String[] args)
59     throws IOException, InterruptedException, ClassNotFoundException {
60
61     Configuration conf = new Configuration();
62     Job job = Job.getInstance(conf, "Binning By Country");
63     job.setJarByClass(BinningByCountry.class);
64     job.setMapperClass(BinMapper.class);
65     job.setNumReduceTasks(0);
66
67     job.setOutputKeyClass(Text.class);
68     job.setOutputValueClass(NullWritable.class);
69
70     TextInputFormat.setInputPaths(job, new Path(args[0]));
71     FileOutputFormat.setOutputPath(job, new Path(args[1]));
72
73     MultipleOutputs.addNamedOutput(job, "bins", TextOutputFormat.class,
74         Text.class, NullWritable.class);
75     MultipleOutputs.setCountersEnabled(job, true);
76
77     System.exit(job.waitForCompletion(true) ? 0 : 1);
78 }
79 }

```

Execution

```
YTRUN-05_binning_by_country

D:\Desktop\bdal_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\
*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib\* -d D:\Desktop\bdal_project\BinningByCo
untry\target\classes D:\Desktop\bdal_project\BinningByCountry\src\main\java\binningbycountry\BinningByCoun
try.java

D:\Desktop\bdal_project>jar -cvf D:\Desktop\bdal_project\BinningByCountry\target\BinningByCountry.jar -C D
:\Desktop\bdal_project\BinningByCountry\target\classes .
added manifest
adding: binningbycountry/(in = 0) (out= 0)(stored 0%)
adding: binningbycountry/BinningByCountry$BinMapper.class(in = 2928) (out= 1073)(deflated 63%)
adding: binningbycountry/BinningByCountry.class(in = 1967) (out= 952)(deflated 51%)

D:\Desktop\bdal_project>hadoop jar D:\Desktop\bdal_project\BinningByCountry\target\BinningByCountry.jar bi
nningbycountry.BinningByCountry /BdalProject/data /BdalProject/output/binningbycountry
2026-09-08 17:50:53,328 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:50:54,470 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perform
ed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:50:54,485 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
-yarn/staging/HP/.staging/job_1788866353307_0010
2026-09-08 17:50:54,737 INFO input.FileInputFormat: Total input files to process : 13
2026-09-08 17:50:54,815 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:50:54,996 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0010
2026-09-08 17:50:54,997 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:50:55,256 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:50:55,257 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:50:55,343 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0010
2026-09-08 17:50:55,394 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/ap
plication_1788866353307_0010/
2026-09-08 17:50:55,397 INFO mapreduce.Job: Running job: job_1788866353307_0010
2026-09-08 17:51:11,709 INFO mapreduce.Job: Job job_1788866353307_0010 running in uber mode : false
2026-09-08 17:51:11,711 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:51:36,370 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:51:37,392 INFO mapreduce.Job: map 15% reduce 0%
2026-09-08 17:51:45,652 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:51:47,720 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:51:48,732 INFO mapreduce.Job: map 38% reduce 0%
2026-09-08 17:51:50,811 INFO mapreduce.Job: map 46% reduce 0%
2026-09-08 17:51:53,910 INFO mapreduce.Job: map 54% reduce 0%
2026-09-08 17:52:11,356 INFO mapreduce.Job: map 62% reduce 0%
2026-09-08 17:52:15,551 INFO mapreduce.Job: map 69% reduce 0%
2026-09-08 17:52:17,593 INFO mapreduce.Job: map 85% reduce 0%
2026-09-08 17:52:20,647 INFO mapreduce.Job: map 100% reduce 0%
```

```
YTRUN-05_binning_by_country
003
-rw-r--r-- 1 HP supergroup 1415 2026-09-08 17:51 /BdalProject/output/binningbycountry/Bahrain-m-00
005
-rw-r--r-- 1 HP supergroup 85 2026-09-08 17:52 /BdalProject/output/binningbycountry/Bahrain-m-00
009
-rw-r--r-- 1 HP supergroup 19336 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
000
-rw-r--r-- 1 HP supergroup 20847 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
001
-rw-r--r-- 1 HP supergroup 14254 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
002
-rw-r--r-- 1 HP supergroup 7468 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
003
-rw-r--r-- 1 HP supergroup 10102 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
004
-rw-r--r-- 1 HP supergroup 11417 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
005
-rw-r--r-- 1 HP supergroup 17793 2026-09-08 17:51 /BdalProject/output/binningbycountry/Belgium-m-00
006
-rw-r--r-- 1 HP supergroup 12752 2026-09-08 17:52 /BdalProject/output/binningbycountry/Belgium-m-00
007
-rw-r--r-- 1 HP supergroup 15256 2026-09-08 17:52 /BdalProject/output/binningbycountry/Belgium-m-00
008
-rw-r--r-- 1 HP supergroup 4876 2026-09-08 17:52 /BdalProject/output/binningbycountry/Belgium-m-00
009
-rw-r--r-- 1 HP supergroup 9564 2026-09-08 17:52 /BdalProject/output/binningbycountry/Belgium-m-00
010
-rw-r--r-- 1 HP supergroup 9224 2026-09-08 17:52 /BdalProject/output/binningbycountry/Belgium-m-00
011
-rw-r--r-- 1 HP supergroup 7553 2026-09-08 17:52 /BdalProject/output/binningbycountry/Belgium-m-00
012
-rw-r--r-- 1 HP supergroup 2461 2026-09-08 17:52 /BdalProject/output/binningbycountry/Brazil-m-000
10
-rw-r--r-- 1 HP supergroup 6057 2026-09-08 17:51 /BdalProject/output/binningbycountry/Canada-m-000
04
-rw-r--r-- 1 HP supergroup 79 2026-09-08 17:51 /BdalProject/output/binningbycountry/Canada-m-000
05
-rw-r--r-- 1 HP supergroup 4391 2026-09-08 17:51 /BdalProject/output/binningbycountry/Canada-m-000
06
-rw-r--r-- 1 HP supergroup 777 2026-09-08 17:52 /BdalProject/output/binningbycountry/Canada-m-000
07
-rw-r--r-- 1 HP supergroup 410 2026-09-08 17:52 /BdalProject/output/binningbycountry/Canada-m-000
08
```

```
YTRUN-05_binning_by_country_out
D:\Desktop\bdal_project>hdfs dfs -ls /BdalProject/output/binningbycountry
Found 328 items
-rw-r--r-- 1 HP supergroup 3639 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00000
-rw-r--r-- 1 HP supergroup 9116 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00001
-rw-r--r-- 1 HP supergroup 8397 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00002
-rw-r--r-- 1 HP supergroup 2829 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00003
-rw-r--r-- 1 HP supergroup 17827 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00004
-rw-r--r-- 1 HP supergroup 9498 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00005
-rw-r--r-- 1 HP supergroup 14036 2026-09-08 17:51 /BdalProject/output/binningbycountry/Australia-m-00006
-rw-r--r-- 1 HP supergroup 8847 2026-09-08 17:52 /BdalProject/output/binningbycountry/Australia-m-00007
-rw-r--r-- 1 HP supergroup 8711 2026-09-08 17:52 /BdalProject/output/binningbycountry/Australia-m-00008
-rw-r--r-- 1 HP supergroup 10042 2026-09-08 17:52 /BdalProject/output/binningbycountry/Australia-m-00009
-rw-r--r-- 1 HP supergroup 1570 2026-09-08 17:52 /BdalProject/output/binningbycountry/Australia-m-00010
-rw-r--r-- 1 HP supergroup 7428 2026-09-08 17:52 /BdalProject/output/binningbycountry/Australia-m-00011
-rw-r--r-- 1 HP supergroup 5556 2026-09-08 17:51 /BdalProject/output/binningbycountry/Austria-m-00000
-rw-r--r-- 1 HP supergroup 4281 2026-09-08 17:51 /BdalProject/output/binningbycountry/Austria-m-00001
-rw-r--r-- 1 HP supergroup 704 2026-09-08 17:51 /BdalProject/output/binningbycountry/Austria-m-00003
-rw-r--r-- 1 HP supergroup 4319 2026-09-08 17:51 /BdalProject/output/binningbycountry/Austria-m-00004
-rw-r--r-- 1 HP supergroup 3981 2026-09-08 17:51 /BdalProject/output/binningbycountry/Austria-m-00005
-rw-r--r-- 1 HP supergroup 163 2026-09-08 17:51 /BdalProject/output/binningbycountry/Austria-m-00006
-rw-r--r-- 1 HP supergroup 1387 2026-09-08 17:52 /BdalProject/output/binningbycountry/Austria-m-00007
-rw-r--r-- 1 HP supergroup 6901 2026-09-08 17:52 /BdalProject/output/binningbycountry/Austria-m-00008
```

YTRUN-05_binning_by_country_out							
10	-rw-r--r--	1	HP supergroup	2162	2026-09-08	17:52	/BdalProject/output/binningbycountry/Poland-m-00011
11	-rw-r--r--	1	HP supergroup	11558	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00000
0000	-rw-r--r--	1	HP supergroup	35319	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00001
0001	-rw-r--r--	1	HP supergroup	6001	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00002
0002	-rw-r--r--	1	HP supergroup	9415	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00003
0003	-rw-r--r--	1	HP supergroup	7152	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00004
0004	-rw-r--r--	1	HP supergroup	6597	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00005
0005	-rw-r--r--	1	HP supergroup	4001	2026-09-08	17:51	/BdalProject/output/binningbycountry/Portugal-m-00006
0006	-rw-r--r--	1	HP supergroup	9040	2026-09-08	17:52	/BdalProject/output/binningbycountry/Portugal-m-00007
0007	-rw-r--r--	1	HP supergroup	3255	2026-09-08	17:52	/BdalProject/output/binningbycountry/Portugal-m-00008
0008	-rw-r--r--	1	HP supergroup	9809	2026-09-08	17:52	/BdalProject/output/binningbycountry/Portugal-m-00009
0009	-rw-r--r--	1	HP supergroup	6281	2026-09-08	17:52	/BdalProject/output/binningbycountry/Portugal-m-00010
0010	-rw-r--r--	1	HP supergroup	4524	2026-09-08	17:52	/BdalProject/output/binningbycountry/Portugal-m-00011
0011	-rw-r--r--	1	HP supergroup	8119	2026-09-08	17:52	/BdalProject/output/binningbycountry/Portugal-m-00012
0012	-rw-r--r--	1	HP supergroup	4190	2026-09-08	17:51	/BdalProject/output/binningbycountry/RSA-m-00001
-m-00007	-rw-r--r--	1	HP supergroup	92	2026-09-08	17:52	/BdalProject/output/binningbycountry/Saudi_Arabia
-m-00011	-rw-r--r--	1	HP supergroup	754	2026-09-08	17:52	/BdalProject/output/binningbycountry/Saudi_Arabia
00001	-rw-r--r--	1	HP supergroup	2831	2026-09-08	17:51	/BdalProject/output/binningbycountry/Singapore-m-00004
00004	-rw-r--r--	1	HP supergroup	6178	2026-09-08	17:51	/BdalProject/output/binningbycountry/Singapore-m-00009
00009	-rw-r--r--	1	HP supergroup	4608	2026-09-08	17:52	/BdalProject/output/binningbycountry/Singapore-m-00010
00010	-rw-r--r--	1	HP supergroup	4764	2026-09-08	17:52	/BdalProject/output/binningbycountry/Singapore-m-00010
-rw-r--r--	-rw-r--r--	1	HP supergroup	24141	2026-09-08	17:51	/BdalProject/output/binningbycountry/Spain-m-00000

Pig Analysis

The Pig scripts were executed in local mode against the same comma separated dataset used by the MapReduce jobs.

1) Top 5 Countries by Revenue

Returns the five countries generating the highest total sales value.

Script

```
1  -- 1) Top 5 Countries by Revenue : returns the five countries generating the
2  -- highest total sales value.
3  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
4  (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
5   invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
6  -- an InvoiceNo starting with C is a cancellation and would subtract from real sales
7  valid = FILTER retail BY country IS NOT NULL AND quantity > 0 AND unitprice > 0
8         AND NOT (invoiceno MATCHES 'C.*');
9  lines = FOREACH valid GENERATE country, (quantity * unitprice) AS revenue;
10 grp_country = GROUP lines BY country;
11 revenue_by_country = FOREACH grp_country GENERATE group AS country,
12                  ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue;
13 sorted_desc = ORDER revenue_by_country BY total_revenue DESC;
14 top5_countries = LIMIT sorted_desc 5;
15 STORE top5_countries INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top5CountriesByRevenue' USING PigStorage('|');
```

Execution

```
YTRUN-06_pig_top5_countries_by_revenue

D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top5_countries_by_revenue.pig
2026-09-08 17:54:04,466 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:54:04,467 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:54:05,176 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:54:05,177 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
445172.log
2026-09-08 17:54:05,207 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:54:05,876 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:54:05,968 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:54:05,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:54:06,006 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top5_countrie
s_by_revenue.pig-8659662c-67b9-4e0c-b8ca-1df75cca6579
2026-09-08 17:54:06,006 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:54:07,051 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:54:07,056 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:54:07,080 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-08 17:54:07,237 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:54:07,321 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:54:07,391 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $4, $6
2026-09-08 17:54:07,521 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:54:07,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:54:07,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-67
2026-09-08 17:54:07,634 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:07,635 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-06_pig_top5_countries_by_revenue
2026-09-08 17:54:07,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOpt
timizerMR - Using Secondary Key Optimization for MapReduce node scope-67
2026-09-08 17:54:07,634 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:07,635 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size after optimization: 4
2026-09-08 17:54:07,743 [main] INFO org.apache.hadoop.metrics2.impl.MetricsConfig - Loaded properties from hadoo
p-metrics2.properties
2026-09-08 17:54:07,849 [main] INFO org.apache.hadoop.metrics2.impl.MetricsSystemImpl - Scheduled Metric snapsh
ot period at 10 second(s).
2026-09-08 17:54:07,850 [main] INFO org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em started
2026-09-08 17:54:07,888 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings
are added to the job
2026-09-08 17:54:07,904 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markre
set.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-08 17:54:07,905 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-08 17:54:07,907 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress i
s deprecated. Instead, use mapreduce.output.fileoutputformat.compress
set.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-08 17:54:07,905 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-08 17:54:07,907 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress i
s deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-08 17:54:07,909 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Reduce phase detected, estimating # of required reducers.
2026-09-08 17:54:07,911 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEst
imator
2026-09-08 17:54:07,955 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReduc
erEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=45458480
2026-09-08 17:54:07,956 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Setting Parallelism to 1
2026-09-08 17:54:07,956 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is d
eprecated. Instead, use mapreduce.job.reduces
2026-09-08 17:54:07,971 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Setting up single store job
2026-09-08 17:54:07,988 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, w
ill not generate code.
2026-09-08 17:54:07,989 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generate
d code to distributed cacche
2026-09-08 17:54:07,989 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Distributed cache not supported o
```

```
YTRUN-06_pig_top5_countries_by_revenue_out
D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\Top5CountriesByRevenue\part-r-0000
0
United Kingdom|9025222.08
Netherlands|285446.34
EIRE|283453.96
Germany|228867.14
France|209715.11
```

2) Top 10 Products by Revenue

Returns the ten products earning the most revenue.

Script

```
1  -- 2) Top 10 Products by Revenue : returns the ten products earning the most.
2  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
3    (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
4     invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
5  valid = FILTER retail BY quantity > 0 AND unitprice > 0
6         AND NOT (invoiceno MATCHES 'C.*');
7  lines = FOREACH valid GENERATE stockcode, description, (quantity * unitprice) AS revenue;
8  grp_product = GROUP lines BY stockcode;
9  revenue_by_product = FOREACH grp_product GENERATE group AS stockcode,
10    MAX(lines.description) AS description,
11    ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue;
12  sorted_desc = ORDER revenue_by_product BY total_revenue DESC;
13  top10_products = LIMIT sorted_desc 10;
14  STORE top10_products INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top10ProductsByRevenue' USING PigStorage('|');
```

Execution

```
YTRUN-07_pig_top10_products_by_revenue

D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top10_products_by_revenue.pig
2026-09-08 17:54:47,300 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:54:47,302 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:54:48,053 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:54:48,053 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
488047.log
2026-09-08 17:54:48,089 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:54:49,157 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\HP/.pigbootup
not found
2026-09-08 17:54:49,368 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:54:49,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:54:49,491 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10_product
s_by_revenue.pig-391161be-b787-492f-afff-5d0df2f029fe
2026-09-08 17:54:49,497 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:54:50,546 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:54:50,552 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:54:50,585 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP BY, ORDER BY, FILTER, LIMIT
2026-09-08 17:54:50,764 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Insert, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInsert]}
2026-09-08 17:54:50,840 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:54:50,924 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $4, $6, $7
2026-09-08 17:54:51,077 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:54:51,151 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:54:51,190 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-72
2026-09-08 17:54:51,215 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:51,215 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-07_pig_top10_products_by_revenue
2026-09-08 17:55:03,343 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set
. User classes may not be found. See Job or Job#setJar(String).
2026-09-08 17:55:03,352 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input f
iles to process : 1
2026-09-08 17:55:03,352 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total
input paths to process : 1
2026-09-08 17:55:03,353 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total
input paths (combined) to process : 1
2026-09-08 17:55:03,362 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-08 17:55:03,386 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job:
job_local1620493607_0003
2026-09-08 17:55:03,386 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-08 17:55:03,511 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://lo
calhost:8080/
2026-09-08 17:55:03,512 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - OutputCommitter set in confi
g null
2026-09-08 17:55:03,521 [Thread-25] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks
is deprecated. Instead, use mapreduce.job.reduces
2026-09-08 17:55:03,522 [Thread-25] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker
is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:55:03,522 [Thread-25] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.m
arkreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-08 17:55:03,523 [Thread-25] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - File Outp
ut Committer Algorithm version is 2
2026-09-08 17:55:03,523 [Thread-25] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutpu
tCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-09-08 17:55:03,523 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - OutputCommitter is org.apach
e.pig.backend.hadoop.executionengine.mapReduceLayer.PigOutputCommitter
2026-09-08 17:55:03,531 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - Waiting for map tasks
2026-09-08 17:55:03,531 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.LocalJobRunner - St
arting task: attempt_local1620493607_0003_m_000000_0
2026-09-08 17:55:03,538 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapreduce.lib.output.FileO
utputCommitter - File Output Committer Algorithm version is 2
2026-09-08 17:55:03,539 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapreduce.lib.output.FileO
utputCommitter - FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanu
p failures: false
2026-09-08 17:55:03,539 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.yarn.util.ProcfsBasedProce
ssTree - ProcfsBasedProcessTree currently is supported only on Linux.
2026-09-08 17:55:03,579 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.Task - Using Resou
rceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBasedProcessTree@38db425
2026-09-08 17:55:03,582 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - Processin
g split: Number of splits :1
Total Length = 200973
```

```
YTRUN-07_pig_top10_products_by_revenue_out
D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\Top10ProductsByRevenue\part-r-0000
0
DOT|DOTCOM POSTAGE|206248.77
22423|REGENCY CAKESTAND 3 TIER|174484.74
23843|PAPER CRAFT LITTLE BIRDIE|168469.6
85123A|WHITE HANGING HEART T-LIGHT HOLDER|104518.8
47566|PARTY BUNTING|99504.33
85099B|JUMBO BAG RED RETROSPOT|94340.05
23166|MEDIUM CERAMIC TOP STORAGE JAR|81700.92
M|Manual|78110.27
POST|POSTAGE|78101.88
23084|RABBIT NIGHT LIGHT|66964.99
```

3) Top 10 Products by Country

Returns, for every country, its ten best selling products by revenue, using a nested FOREACH so the ordering and limiting happen inside each group.

Script

```
1  -- 3) Top 10 Products by Country : returns, for every country, its ten
2  --    best-selling products by revenue. Uses a nested FOREACH so the ordering
3  --    and limiting happen inside each country's group.
4  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
5    (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
6     invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
7  valid = FILTER retail BY country IS NOT NULL AND quantity > 0 AND unitprice > 0
8         AND NOT (invoiceno MATCHES 'C.*');
9  lines = FOREACH valid GENERATE country, stockcode, (quantity * unitprice) AS revenue;
10 grp_cp = GROUP lines BY (country, stockcode);
11 product_totals = FOREACH grp_cp GENERATE group.country AS country,
12        group.stockcode AS stockcode,
13        ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue;
14 grp_country = GROUP product_totals BY country;
15 top10_per_country = FOREACH grp_country {
16     sorted = ORDER product_totals BY total_revenue DESC;
17     top10 = LIMIT sorted 10;
18     GENERATE FLATTEN(top10);
19 };
20 STORE top10_per_country INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top10ProductsByCountry' USING PigStorage('|');
```

Execution

```
YTRUN-08_pig_top10_products_by_country
D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top10_products_by_country.pig
2026-09-08 17:55:52,258 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:55:52,258 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:55:52,912 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:55:52,912 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
552906.log
2026-09-08 17:55:52,933 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:55:53,522 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:55:53,589 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:55:53,591 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:55:53,627 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10_product
s_by_country.pig-0ea49196-15e6-4d20-a87c-12a782cb361d
2026-09-08 17:55:53,627 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:55:54,303 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:55:54,306 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:55:54,325 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,FILTER
2026-09-08 17:55:54,430 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:55:54,470 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:55:54,524 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $2, $4, $6
2026-09-08 17:55:54,596 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:55:54,645 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:55:54,670 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 2
2026-09-08 17:55:54,670 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size after optimization: 2
2026-09-08 17:55:54,769 [main] INFO org.apache.hadoop.metrics2.impl.MetricsConfig - Loaded properties from hado
```

```
YTRUN-08_pig_top10_products_by_country
2026-09-08 17:56:03,128 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - Merging 1 files, 660910 bytes from disk
2026-09-08 17:56:03,128 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - Merging 0 segments, 0 bytes from memory into reduce
2026-09-08 17:56:03,128 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Merger - Merging 1 sorted segments
2026-09-08 17:56:03,131 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Merger - Down to the last merge-pass, with 1 segments left of total size: 660898 bytes
2026-09-08 17:56:03,131 [pool-9-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - 1 / 1 copied.
2026-09-08 17:56:03,134 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter Algorithm version is 2
2026-09-08 17:56:03,134 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter skip cleanup temporary folders under output directory:false, ignore cleanup failures: false
2026-09-08 17:56:03,140 [pool-9-thread-1] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:56:03,142 [pool-9-thread-1] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-08 17:56:03,146 [pool-9-thread-1] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.PigMapReduce$Reduce - Aliases being processed per job phase (AliasName[line,offset]): M: grp_country[14,14] C: R: top10_per_country[15,20],sorted[16,33]
2026-09-08 17:56:03,357 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Task - Task:attempt_local1250976658_0002_r_000000_0 is done. And is in the process of committing
2026-09-08 17:56:03,363 [pool-9-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - 1 / 1 copied.
2026-09-08 17:56:03,365 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Task - Task:attempt_local1250976658_0002_r_000000_0 is allowed to commit now
2026-09-08 17:56:03,374 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - Saved output of task 'attempt_local1250976658_0002_r_000000_0' to file:/D:/Desktop/bdal_project/PigAnalysis/output/Top10ProductsByCountry
2026-09-08 17:56:03,376 [pool-9-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - reduce > reduce
2026-09-08 17:56:03,376 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Task - Task:attempt_local1250976658_0002_r_000000_0 done.
2026-09-08 17:56:03,377 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Task - Final Counters for attempt_local1250976658_0002_r_000000_0: Counters: 24
    File System Counters
        FILE: Number of bytes read=49535401
        FILE: Number of bytes written=5263942
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
    Map-Reduce Framework
        Combine input records=0
        Combine output records=0
        Reduce input groups=38
        Reduce shuffle bytes=660910
```

```
YTRUN-08_pig_top10_products_by_country_out
D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\Top10ProductsByCountry\part-r-0000
0
RSA|21340|38.25
RSA|22605|29.9
RSA|23298|29.7
RSA|22423|25.5
RSA|22526|25.5
RSA|85066|25.5
RSA|22960|25.5
RSA|84375|25.2
RSA|23240|24.9
RSA|23469|24.9
USA|23328|162.72
USA|22423|114.75
USA|21121|90.0
USA|21124|90.0
USA|21122|90.0
USA|21123|90.0
USA|23158|74.88
USA|23366|57.2
USA|23392|49.92
USA|20975|46.8
EIRE|22423|7844.25
EIRE|M|7049.66
EIRE|C2|5240.0
EIRE|22838|4265.55
EIRE|22960|3097.5
EIRE|85123A|2928.6
EIRE|21843|2781.3
EIRE|22699|2620.8
EIRE|20914|2469.4
EIRE|21166|2304.08
Italy|POST|1663.0
Italy|22720|252.45
Italy|22847|247.2
Italy|22139|222.75
Italy|22960|217.5
Italy|22326|177.0
Italy|22844|159.8
Italy|85123A|159.3
Italy|22939|158.4
Italy|22120|149.25
```

```
YTRUN-08_pig_top10_products_by_country_out
United Arab Emirates|85150|61.2
United Arab Emirates|22139|59.4
United Arab Emirates|85215|46.8
United Arab Emirates|POST|37.5
```

4) Top 10 Customers by Spending

Returns the ten customers who spent the most, excluding rows without a CustomerID.

Script

```
1  -- 4) Top 10 Customers by Spending : returns the ten customers who spent most.
2  -- Rows without a CustomerID are anonymous till sales and cannot be attributed.
3  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
4  (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
5   invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
6  valid = FILTER retail BY customerid IS NOT NULL AND customerid != ''
7          AND quantity > 0 AND unitprice > 0
8          AND NOT (invoiceno MATCHES 'C.*');
9  lines = FOREACH valid GENERATE customerid, country, (quantity * unitprice) AS revenue;
10 grp_customer = GROUP lines BY customerid;
11 spend_by_customer = FOREACH grp_customer GENERATE group AS customerid,
12                     MAX(lines.country) AS country,
13                     ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_spend;
14 sorted_desc = ORDER spend_by_customer BY total_spend DESC;
15 top10_customers = LIMIT sorted_desc 10;
16 STORE top10_customers INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top10CustomersBySpending' USING PigStorage('|');
```

Execution

```
YTRUN-09_pig_top10_customers_by_spending
D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top10_customers_by_spending.pig
2026-09-08 17:56:49,120 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:56:49,121 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:56:49,784 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:56:49,784 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
609780.log
2026-09-08 17:56:49,808 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:56:50,559 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:56:50,624 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:56:50,627 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:56:50,660 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10_custome
rs_by_spending.pig-9ec207a4-bf4d-4d07-a9f6-20756e3cdeec
2026-09-08 17:56:50,661 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:56:51,341 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:56:51,344 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:56:51,365 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-08 17:56:51,477 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Insertion, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInsertion]}
2026-09-08 17:56:51,518 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:56:51,581 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $4
2026-09-08 17:56:51,672 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:56:51,726 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:56:51,748 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-80
2026-09-08 17:56:51,758 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:56:51,758 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-09_pig_top10_customers_by_spending
2026-09-08 17:57:00,195 [JobControl] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-08 17:57:00,202 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-08 17:57:00,214 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-08 17:57:00,214 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-08 17:57:00,214 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-08 17:57:00,227 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-08 17:57:00,253 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_local1781930435_0003
2026-09-08 17:57:00,253 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-08 17:57:00,348 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://localhost:8080/
2026-09-08 17:57:00,350 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - OutputCommitter set in config null
2026-09-08 17:57:00,355 [Thread-25] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-08 17:57:00,356 [Thread-25] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:57:00,356 [Thread-25] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-08 17:57:00,357 [Thread-25] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - File OutputCommitter Algorithm version is 2
2026-09-08 17:57:00,358 [Thread-25] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-09-08 17:57:00,367 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.LocalJobRunner - OutputCommitter is org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.PigOutputCommitter
2026-09-08 17:57:00,366 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - Waiting for map tasks
2026-09-08 17:57:00,367 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.LocalJobRunner - Starting task: attempt_local1781930435_0003_m_000000_0
2026-09-08 17:57:00,383 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - File OutputCommitter Algorithm version is 2
2026-09-08 17:57:00,383 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-09-08 17:57:00,385 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.yarn.util.ProcfsBasedProcessTree - ProcfsBasedProcessTree currently is supported only on Linux.
2026-09-08 17:57:00,433 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.Task - Using ResourceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBasedProcessTree@1e7aaecd
2026-09-08 17:57:00,436 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - Processing
```

```
YTRUN-09_pig_top10_customers_by_spending_out
D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\Top10CustomersBySpending\part-r-000000
14646|Netherlands|280206.02
18102|United Kingdom|259657.3
17450|United Kingdom|194550.79
16446|United Kingdom|168472.5
14911|EIRE|143825.06
12415|Australia|124914.53
14156|EIRE|117379.63
17511|United Kingdom|91062.38
16029|United Kingdom|81024.84
12346|United Kingdom|77183.6
```

5) Monthly Sales Trend

Returns the total revenue, the number of order lines and the number of distinct invoices for each month of trading.

Script

```
1  -- 5) Monthly Sales Trend : returns total revenue, order-line count and number
2  --   of distinct invoices for each month of trading.
3  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
4    (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
5     invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
6  valid = FILTER retail BY quantity > 0 AND unitprice > 0
7    AND NOT (invoiceno MATCHES 'C.*');
8  -- InvoiceDate is 'YYYY-MM-DD HH:MM', so the first seven characters are the month
9  lines = FOREACH valid GENERATE SUBSTRING(invoicedate, 0, 7) AS month,
10    invoiceno,
11    (quantity * unitprice) AS revenue;
12  grp_month = GROUP lines BY month;
13  monthly = FOREACH grp_month {
14    invoices = DISTINCT lines.invoiceno;
15    GENERATE group AS month,
16      ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue,
17      COUNT(lines) AS order_lines,
18      COUNT(invoices) AS invoice_count;
19  };
20  monthly_sorted = ORDER monthly BY month ASC;
21  STORE monthly_sorted INTO 'D:/Desktop/bdal_project/PigAnalysis/output/MonthlySalesTrend' USING PigStorage('|');
```

Execution

```
YTRUN-10_pig_monthly_sales_trend

D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\monthly_sales_trend.pig
2026-09-08 17:57:43,965 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:57:43,966 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:57:44,597 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:57:44,597 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
664592.log
2026-09-08 17:57:44,621 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:57:45,296 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:57:45,372 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:57:45,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:57:45,417 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-monthly_sales
_trend.pig-a0c243d1-e2d8-40d0-a636-cf88dce2b935
2026-09-08 17:57:45,417 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:57:46,152 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:57:46,158 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:57:46,188 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,ORDER_BY,FILTER
2026-09-08 17:57:46,292 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Insertor, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInsertor]}
2026-09-08 17:57:46,330 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:57:46,378 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $6, $7
2026-09-08 17:57:46,461 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:57:46,554 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:57:46,575 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-78
2026-09-08 17:57:46,586 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 3
2026-09-08 17:57:46,586 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

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YTRUN-10_pig_monthly_sales_trend

e Output Committer Algorithm version is 2
2026-09-08 17:57:55,402 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-09-08 17:57:55,405 [pool-9-thread-1] INFO org.apache.hadoop.yarn.util.ProcfsBasedProcessTree - ProcfsBasedProcessTree currently is supported only on Linux.
2026-09-08 17:57:55,445 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Task - Using ResourceCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBasedProcessTree@4fea39c1
2026-09-08 17:57:55,445 [pool-9-thread-1] INFO org.apache.hadoop.mapred.ReduceTask - Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@10ea0c1f
2026-09-08 17:57:55,446 [pool-9-thread-1] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-08 17:57:55,449 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - MergeManager: memoryLimit=652528832, maxSingleShuffleLimit=163132208, mergeThreshold=430669056, ioSortFactor=10, memoryToMergeOutputsThreshold=10
2026-09-08 17:57:55,565 [EventFetcher for fetching Map Completion Events] INFO org.apache.hadoop.mapreduce.task.reduce.EventFetcher - attempt_local643206064_0002_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-09-08 17:57:55,573 [localfetcher#2] INFO org.apache.hadoop.mapreduce.task.reduce.LocalFetcher - localfetcher#2 about to shuffle output of map attempt_local643206064_0002_m_000000_0 decomp: 444 len: 448 to MEMORY
2026-09-08 17:57:55,575 [localfetcher#2] INFO org.apache.hadoop.mapreduce.task.reduce.InMemoryMapOutput - Read 444 bytes from map-output for attempt_local643206064_0002_m_000000_0
2026-09-08 17:57:55,577 [localfetcher#2] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - closeInMemoryFile -> map-output of size: 444, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory -> 444
2026-09-08 17:57:55,578 [EventFetcher for fetching Map Completion Events] INFO org.apache.hadoop.mapreduce.task.reduce.EventFetcher - EventFetcher is interrupted.. Returning
2026-09-08 17:57:55,580 [pool-9-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - 1 / 1 copied.
2026-09-08 17:57:55,580 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - final Merge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-09-08 17:57:55,590 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Merger - Merging 1 sorted segments
2026-09-08 17:57:55,590 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Merger - Down to the last merge-pass, with 1 segments left of total size: 423 bytes
2026-09-08 17:57:55,593 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - Merged 1 segments, 444 bytes to disk to satisfy reduce memory limit
2026-09-08 17:57:55,595 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - Merging 1 files, 448 bytes from disk
2026-09-08 17:57:55,596 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.task.reduce.MergeManagerImpl - Merging 0 segments, 0 bytes from memory into reduce
2026-09-08 17:57:55,596 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Merger - Merging 1 sorted segments
2026-09-08 17:57:55,597 [pool-9-thread-1] INFO org.apache.hadoop.mapred.Merger - Down to the last merge-pass, with 1 segments left of total size: 423 bytes
2026-09-08 17:57:55,598 [pool-9-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - 1 / 1 copied.
2026-09-08 17:57:55,602 [pool-9-thread-1] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutput Committer Algorithm version is 2
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YTRUN-10_pig_monthly_sales_trend_out

D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\MonthlySalesTrend\part-r-00000
2010-12|823746.14|41480|1559
2011-01|691364.56|34306|1086
2011-02|523631.89|27105|1100
2011-03|717639.36|35803|1454
2011-04|537808.62|29096|1246
2011-05|770536.02|36164|1681
2011-06|761739.9|35977|1533
2011-07|719221.19|38645|1475
2011-08|759138.38|34483|1361
2011-09|1058590.17|49261|1837
2011-10|1154979.3|59304|2040
2011-11|1509496.33|83369|2769
2011-12|638792.68|25111|819
```